

Jaime A. Lopez-Merizalde, Ph.D.

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Professional Summary

STEM professional with 5+ years of experience across data science, software engineering, and applied mathematics. Proficient in using a variety of software tools to derive data insights and perform scientific computations. This includes leveraging Python, C, SQL, etc. Comfortable navigating technical challenges and communicating relevant nuances to a non-technical audience. Capable of working in technical groups, participating in collaborations, and actively furthering team goals and outcomes.

Core Competencies

- Python, Jupyter, C, ADPL, Pandas, Sklearn, Numpy, SQL, Git, JAVA, Matlab, R, Snowflake, ANSYS
- Data Pipelines and Data Analysis
- Applied High-Performance Computation
- Scientific Communication
- Cross-functional Collaboration

Professional Experience

U.S. Food & Drug Administration (FDA)

ORISE Mathematics Postdoctoral Research Fellow

Washington, D.C. (Remote)

October 2023 - Present

- Build technical workflow pipelines from scratch using BASH and Python, resulting in efficient and repeatable computational routines.
- Develop computational fluid dynamics (CFD) models for research on Ethylene Oxide alternatives leading to publications and proven sterilization techniques.
- Write platform-specific configurations in ANSYS Fluent to carry out computations on high-performance computing (HPC) platforms resulting in valuable datasets for critical evaluation of sterilization efficacy.
- Collect, cleanse, analyze, and interpret large data sets (on the order of 100GBs).
- Engage with stakeholders to gather project requirements and follow-up with live experimental validations.
- Present and communicate computational and laboratory research outcomes to non-technical audiences answering questions in a digestible manner emphasizing key take-aways.
- Collaborate with leading engineers and scientists to develop necessary validation experiments.
- Built a research science tool to predict the dose of Vaporized Hydrogen Peroxide in lumens (<https://huggingface.co/spaces/osel-dbcms/VaporSterilizationSuite>).
- Set up and manage experimental laboratory validations for computational results, leading to increased confidence in the models developed.

Pet Desk

Data Science Engineering Contractor

San Diego, CA (Remote)

May 2022 - August 2023

- Developed data reporting infrastructure in Python using Jupyter notebooks, Python scripts, R, and SQL reducing manual workload for senior data scientists and improving overall backend infrastructure.
- Built connection processes and libraries for data analytics connecting Snowflake and Google's developer platform allowing faster deployment to C-suite reports.
- Created automation pipelines and scalable scripts from end-to-end for recurring reports using Buildkite's platform.
- Created in-house automated quality control for reporting pipelines leading to uncovering multiple faults for the BI and customer success teams.
- Performed ad-hoc reporting as needed in response to critical client requests ensuring service reliability.
- Participated in code reviews and led early-entry software developers in training sessions for the platform's tools.
- Communicated with non-technical product managers to anticipate and plan for client needs.

21 Sustainability

Backend Developer

New Orleans, LA (Remote)

June 2021 - March 2022

- Developed backend code in Python and PostgreSQL to support a data pipeline using Heroku PaaS.
- Collaborated with a cross-functional team of sociologists, ecologists, and biologists to meet stakeholder requirements.
- Worked independently on self-directed tasks, while supporting the team with database blockers.

Sharpest Minds

Machine Learning Intern

New Orleans, LA (Remote)

March 2021 - March 2022

- Developed a full-stack application to launch a user-facing widget from a Jupyter notebook to support mathematics OCR rendering for LaTeX. This enhances notebooks for users to take math notes more naturally.
- Created the frontend application relying on the popular tkinter library. Focused on user experience and ease of navigation for reduced friction.
- Utilized machine learning engineering approaches and employed them in production-level code to create a pip-installable Python package.
- Worked with an experienced mentor in the machine learning space to develop a user-focused open-source product.
- Built and maintained version-controlled source code with documentation and working instructions for users.

Tulane University
Adjunct Professor/Instructor

New Orleans, LA
August 2018 - January 2021

- Courses taught: Statistics for Scientists, Calculus, Linear Algebra, and Differential Equations.
- Managed teaching assistants to coordinate academic work and taught classes for 30 - 100 students per semester.
- Created original teaching content for online-offline hybrid learning environments.

General Electric - Power
Software Development Intern

New Orleans, LA
May 2019 - May 2020

- Wrote backend software in Java updating or adding processes across GE's microservices affecting over 3000 users.
- Performed PRs for senior software developers, accelerating deployment timelines.
- Developed an internally-facing landing page for admin users for backend available features, easing manual workload for L3 support engineers.
- Used Angular to develop some front end features for the French Quarter Fest App, including the implementation of map-based food and artist search features.
- Provided Spanish translations supporting upcoming product launches in Latin America.

Los Alamos National Laboratory (LANL)
Mathematics Intern: Theoretical Division

Los Alamos, NM
May 2017 - August 2017

- Produced optimized surrogate mathematical models for flow and transport (FT) in rock fractures. This work extended the application of Darcy and Brinkman equations for FT.
- Used LANL's Model Analysis and Decision Support (MADS) high-performance computational framework to train a graph-based model on data created by a high-fidelity, finite element model. This work reduced computation time, making it feasible to run thousands of simulations necessary for uncertainty quantification.
- Developed software using the Julia platform for scientific computing, finding and validating research ideas. This led to scientific insights on flow-channeling effects in low-porosity media and its connection to optimized graph models.
- Collaborated with mathematicians, engineers, physicists, and software engineers to break through technical challenges.
- Reviewed mathematical literature on the needs of hydraulic fracturing in the United States' energy sector, the state-of-the-art research on hydraulic fracture modeling, and proper avenues for solving non-linear inverse problems using the Levenberg-Marquardt Algorithm via MADS.

Education

Tulane University
Doctor of Philosophy (Ph.D.) in Mathematics

New Orleans, LA
2013 - 2020

Dissertation Title: "[Optimized Reduced Models for Discrete Fracture Networks Used in Modeling Particle Flow and Transport.](#)"

Advisor: Dr. James Macklin Hyman.

University of Nebraska
Bachelor of Science (B.S.) in Civil Engineering

Omaha, NE
2006 - 2012

University of Nebraska
Bachelor of Science (B.S.) in Architectural Engineering

Omaha, NE
2006 - 2012

Relevant Courses:

- Survey of Optimization Algorithms • Multidimensional Optimization • Numerical Optimization • Small Sample Statistics • Scientific Computing • Linear Models • Quantifying Uncertainty • Artificial Intelligence • Applied Mathematics • Partial Differential Equations • Epidemiological Mathematical Modeling

Upcoming Publications

Lopez-Merizalde, J., Craven, B.A., & Myers, M. (2025). Methodology for Quantifying Compressibility-Enhanced Sterilant Transport in Medical-Device Lumens, with Application to Sterilization by Vaporized Hydrogen Peroxide. Manuscript in preparation.

Presentations

- Lopez-Merizalde, J., Craven, B.A., & Myers, M. (2025). Methodology for Quantifying Compressibility-Enhanced Sterilant Transport in Medical-Device Lumens, with Application to Sterilization by Vaporized Hydrogen Peroxide. [Poster]. Food and Drug Administration Scientific Symposium 2025.
- Lopez-Merizalde, J., Armando, I., Liu, Y., Saylor, D., Weeks, J. (2025). Regulatory Science Tools for Terminal Sterilization Based on VH₂O₂ Mass Transport Models. [PowerPoint Slides]. Food and Drug Administration.

Awards

- Outstanding Mathematics Dissertation (May 2020): "[Optimized Reduced Models for Discrete Fracture Networks Used in Modeling Particle Flow and Transport.](#)"
- School of Science and Engineering Research Poster Competition 1st place (May 2018): "Modeling Discrete Fracture Networks using Surrogate Graph Networks for Fluid Flow."